

Total No. of Questions : 8]

SEAT No. :

PE2751

[6583]-335

[Total No. of Pages : 2

T.E. (Information Technology)
ADVANCED DATABASE MANAGEMENT SYSTEM
(2019 Pattern) (Semester - V) (314445 B) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1) a)** What is Cassandra? Draw and explain Cassandra Architecture. **[9]**
- b)** What is JSON? How to create object and array object in JSON? Explain with example. **[9]**

OR

- Q2) a)** What is NOSQL database? Explain different types of NOSQL databases with example. **[10]**
- b)** Write a short note on **[8]**
- i) MongoDB Database
 - ii) XML

- Q3) a)** What is data warehousing? Draw and explain data warehouse architecture. **[9]**
- b)** Explain different analytical operations of OLAP. **[8]**

OR

- Q4) a)** Define OLAP. What are different types of OLAP? Explain in detail. **[9]**
- b)** Explain star and snowflake schemas of data warehouse with example. **[8]**

- Q5) a)** What is KDD? Explain knowledge discovery process in detail. **[9]**
- b)** What is data mining. What are different applications of data mining? **[9]**

OR

P.T.O.

- Q6)** a) Define data mining. Explain data mining tasks with suitable example. [10]
b) Explain architecture of data mining system. [8]

- Q7)** a) Write a short note on [10]
i) Spatial Databases
ii) Temporal Databases
b) Draw and explain the architecture of mobile database. [7]

OR

- Q8)** a) Write a short note on [10]
i) Geographical Information Systems
ii) Genome data management
b) Draw and explain Hive architecture [7]

